

A Variational Eigenvalue Solver on a Photonic Quantum Processor

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1. ABSTRACT & THEORETICAL FOUNDATION

We demonstrate a hybrid quantum-classical algorithm using parameterized quantum states and classical parameter optimization to calculate the ground state molecular energy curve of Helium hydride HeH^+ .

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"A team effort: the quantum computer tests a molecule's electron arrangement, while a classical supercomputer tweaks the angles like knobs until it finds the lowest energy state."

3. Mathematical Formulation & Unitary Rule

$$E_0 \leq \langle \psi(\vec{\theta}) | H | \psi(\vec{\theta}) \rangle = \sum_k c_k \langle \theta | U$$

Rayleigh-Ritz variational bound guarantees that any parameterized quantum trial state has energy strictly greater than or equal to ground state E_0 .

4. Step-by-Step Circuit Sequence

Parameterized ansatz circuit $U(\theta)$ followed by basis change rotations for Pauli terms P_k .

5. Executable IBM Qiskit (Python) Code

```
from qiskit.circuit.library import TwoLocal
from qiskit_algorithms import VQE
from qiskit_algorithms.optimizers import COBYLA
from qiskit.primitives import Estimator

# Hardware efficient ansatz: Ry + Rz with CZ entanglers
ansatz = TwoLocal(num_qubits=2, rotation_blocks=['ry', 'rz'], entanglement_blocks='cz')
optimizer = COBYLA(maxiter=100)
estimator = Estimator()
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

$O(\exp(N))$ for exact diagonalization $\rightarrow$ Polynomial gate depth $O(\text{poly}(N))$ Heuristic + classical optimization loop

7. Key Applications & Future Research Directions

- Molecular ground state energy and chemical reaction pathway modeling (drug discovery, battery materials)
- Condensed matter physics and lattice model simulation
- Quantum finance portfolio optimization via quadratic unconstrained models

Future Work:

Hardware-efficient ansatz designs, quantum error mitigation, and scaling to strongly correlated catalyst molecules.